Fiscal Year:	FY 2012	**Task Last Updated:**	FY 02/06/2012
PI Name:	Nickerson, Cheryl A Ph.D.		
Project Title:	RNA Binding Proteins as Evolutionarily Conserved Cellular Spaceflight Response Mechanisms		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	(1) Cell Culture		
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Zip Code:	85287-5401	**Congressional District:**	9
Comments:	NOTE PI moved from Tulane University to Arizona State University in 2006.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2008 Space Biology--Microbial, Plant, Cell Biology: NNH08ZTT003N
Start Date:	05/01/2009	**End Date:**	04/30/2013
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
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Flight Program:	Shuttle/ISS		
Flight Assignment:	STS-131 NOTE: extended to 4/30/2013 per NSSC and A.Chu/ARC (Ed., 4/23/12)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Ott, C. Mark (NASA Johnson Space Center) Wilson, James (Villanova University)		
Grant/Contract No.:	NNX09AH40G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	EDITOR'S NOTE: Ground and Flight portions combined into one project per Ames Research Center (July 2010) Understanding infectious disease risks during spaceflight is critical to provide safe passage for human exploration to the moon and Mars. This issue is especially important given reports that the crew's immune system is dysfunctional during flight, and our recent discovery that spaceflight increased the disease-causing potential and globally altered the gene expression of the human food-borne pathogen Salmonella typhimurium. Our research also showed that many of the observed changes in Salmonella were associated with specific proteins in the cells, called RNA binding proteins, that bind to and regulate genetic material within the cell to control cellular characteristics in response to stress. In addition, our studies during spaceflight and using ground-based, spaceflight analogue culture systems have shown that increasing the concentration of certain ionic salts, like phosphates, in the growth media can prevent the increased disease causing potential observed for Salmonella during spaceflight. Interestingly, human cells also contain evolutionarily conserved RNA binding proteins that are very similar in structure and function to those found in bacteria. In human cells, these RNA binding proteins are critical for the cells proper function, including supporting a healthy immune system, maintaining tissue differentiation, and responding to cellular stress. We hypothesize that RNA binding regulatory proteins and their small RNA binding counterparts are key to a conserved, common cellular spaceflight response mechanism in both bacterial and human cells and that this response can be manipulated by environmental salt/ion levels. The implications of a universally conserved molecular response to spaceflight would affect NASA's approach to infectious disease risk assessment, development of biological processing systems for exploration, and changes in biodegradation during a mission. Indeed, if these responses are also observed in human cells, they may translate into knowledge of both the astronaut's response to infection and possibly larger physiological responses that are influenced by cellular changes.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research will enrich life on Earth through the use of space technology and the application of biomedical knowledge. Specifically, this study will utilize NASA ground-based biotechnology 1) to broaden our knowledge of the host-pathogen interaction between mammalian and microbial cells that leads to infectious disease, and 2) for the development of new therapeutic strategies to combat infectious disease for the general public.
Task Progress:	Previously published results from our lab showed that Salmonella grown in rich LB media in spaceflight and spaceflight analogue conditions (low shear modeled microgravity/LSMMG) displayed increased virulence and global alterations in global gene expression. A central role was identified for the evolutionarily conserved, small regulatory RNA-binding protein Hfq in regulating key aspects of the Salmonella spaceflight and LSMMG response. Interestingly, Salmonella grown in minimal M9 media during spaceflight did not exhibit an increase in virulence, but did demonstrate differential expression of many genes, including those associated with either the regulation of, or regulation by, the Hfq protein and small regulatory RNAs. Moreover, we found that environmental ion concentrations mitigated the enhanced virulence observed in spaceflight-grown Salmonella, and also altered its LSMMG-induced acid stress pathogenesis related response. This led us to hypothesize that RNA binding regulatory proteins and their small RNA binding counterparts are key to a common conserved cellular spaceflight and LSMMG response mechanism. In support of this hypothesis, a recently published joint paper between our lab and that of our collaborators has shown the involvement of Hfq in regulating the LSMMG response of the opportunistic pathogen Pseudomonas aeruginosa (Crabbe et al, 2010, Environ Microbiol). The gene coding for Hfq and several genes belonging to the Hfq regulon were differentially regulated when the transcriptional profiles of P. aeruginosa PAO1 grown in LSMMG and the normal orientation control were compared. Specifically, this study showed for the first time that cultivation of P. aeruginosa PAO1 in the LSMMG environment of the RWV induces a transcriptomic and physiologic response with possible implications for virulence. AlgU was identified as one of the key regulators in the LSMMG-induced response and this alternative sigma factor presumably up-regulated the transcription of genes downstream involved in environmental stress resistance, alginate production and microaerophilic/anaerobic metabolism. Moreover, AlgU might control hfq transcription in P. aeruginosa PAO1, which has previously been linked with Salmonella response to space- and LSMMG conditions. The observations during the present study in combination with a decreased immune responsiveness of astronauts could become problematic during long-term spaceflight missions and should direct future research regarding the risk assessment and prevention of infections caused by this opportunistic pathogen. Moreover, recent work from our lab characterized the spaceflight-induced global transcriptional and proteomic responses of P. aeruginosa. Specifically, P. aeruginosa responded to spaceflight conditions through differential regulation of 167 genes (including pathogenesis-related genes) and 28 proteins, with Hfq as a global transcriptional regulator. Since Hfq was also differentially regulated in spaceflight-grown Salmonella, Hfq represents the first spaceflight-induced regulator acting across bacterial species. In addition, a recent publication from our collaborative team with Staphylococcus aureus reported molecular genetic responses to LSMMG culture reflective of those previously reported for Salmonella and Pseudomonas, including 1) a role for Hfq as a central global regulator of the LSMMG-induced transcriptomic response, 2) LSMMG modulation of the stress response, and 3) LSMMG modulation of pathogenesis-related phenotypes. FLIGHT EXPERIMENT: A NASA-funded flight experiment on STS-131, operationally called STL-IMMUNE, was awarded as a part of this ground-based flight analogue research. The objectives of this flight experiment were directly related to and complemented our ground-based work (including use of the same cell types) the goal of which is to identify universally conserved molecular responses to ground-based spaceflight analogue culture across a variety of bacterial and mammalian cell types. Specifically, the focus of STL-IMMUNE was to 1) understand the effect of microgravity on the host-pathogen interaction and infection process from the perspective of the human cell, and 2) understand the effect of microgravity alone on human cellular responses that are relevant to normal cellular homeostasis or transition to disease. STL-IMMUNE was the first fundamental space biology experiment to conduct an in-flight infection of human cells. We are currently finalizing sample analysis for alterations in protein expression (via global proteomic response profiling/iTRAQ), cellular differentiation, immune function (inflammatory response mediators) and stress responses associated with spaceflight in the presence and absence of infection (all flight sample comparisons were made to identical synchronous ground controls housed in duplicate CCM hardware as that used for flight). To date, we have optimized protein extraction, concentration/purification, and separation methods critical for proteomic analysis via iTRAQ and subsequently used this quantitative mass spectrometry based approach to identify 86 total unique human

	proteins (95% confidence, 1% FDR) that are expressed by the HT-29 intestinal epithelial cells cultured on orbit either before or after infection with Salmonella Typhimurium. These proteins belonged to a range of different functional classes and categories, including those important for metabolism, apoptosis, cell communication, immune function, and others. Of the 89 proteins identified, a subset of these were found to be differentially regulated in response to spaceflight, and showed differences in response to infection between flight and ground cultures. We are currently finalizing global proteomic response profiling for the A549 lung epithelial cells. Moreover, immunohistochemical profiling has revealed intriguing differences in both distribution and expression patterns of targeted host cell proteins and alterations in cell cytoarchitecture.
Bibliography Type:	Description: (Last Updated: 07/02/2025)
Articles in Peer-reviewed Journals	Crabbe A, Sarker SF, Van Houdt R, Ott CM, Leys N, Cornelis P, Nickerson CA. "Alveolar epithelium protects macrophages from quorum sensing-induced cytotoxicity in a three-dimensional co-culture model." Cell Microbiol. 2011 Mar;13(3):469-81. Epub 2010 Nov 25. http://dx.doi.org/10.1111/j.1462-5822.2010.01548.x ; PubMed PMID: 21054742 , Mar-2011
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Awards	Nickerson C. "NASA Exceptional Scientific Achievement Medal, August, 2011." Aug-2011
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Significant Media Coverage	Lewis B. "'ASU spaceflight research pioneer receives top NASA science award.' Article about Dr. Nickerson's NASA Award." ASU News [science & tech], August 4, 2011. https://asunews.asu.edu/20110804_Nickerson , Aug-2011
Significant Media Coverage	Harrington JD, Caspermeyer J. "'Spacebound Bacteria Inspire Earthbound Remedies.' NASA Shuttle/Station News Feature about Dr. Nickerson's flight experiment." NASA Station & Shuttle News, March 21, 2011. http://www.nasa.gov/topics/shuttle_station/features/pseudomonas.html , Mar-2011